

Main Experiment Tables with Local Reproduction Rows

Reproduction summary

1 PruneVid

Method	Retained	FLOPs	MVBench	VideoMME	EgoSchema (Subset / Fullset)
PLLaVA	100.0%	1.00×	46.6	44.4	47.8 / 42.6
PLLaVA w/ FastV	30.0%	0.33×	46.1	43.6	46.2 / 41.0
PLLaVA w/ Prunmerge	55.7%	0.53×	45.6	43.8	45.2 / 40.4
PLLaVA w/ Look-M	20.0%	1.00×	46.6	44.3	47.0 / 42.3
PLLaVA w/ Ours	16.2%	0.23×	47.6	45.0	49.0 / 42.6
PLLaVA w/ Ours (Reproduce)	16.2%	0.23×	46.42	44.89	48.40 / 41.90
ST-LLM	100.0%	1.00×	54.9	42.0	56.2 / 45.6
ST-LLM w/ FastV	30.0%	0.37×	42.9	34.5	48.0 / 38.5
ST-LLM w/ Look-M	20.0%	1.00×	54.0	40.6	54.0 / 44.5
ST-LLM w/ Ours	15.1%	0.26×	54.3	41.4	54.6 / 44.7
ST-LLM w/ Ours (Reproduce, self-port)	15.1%	0.26×	54.80	42.59	59.80 / 45.10
LLaVA-OneVision	100.0%	1.00×	58.0	58.2	62.0 / 60.0
LLaVA-OneVision w/ FastV	30.0%	0.30×	57.2	57.6	62.6 / 60.0
LLaVA-OneVision w/ Prunmerge	55.2%	0.49×	52.9	56.7	62.2 / 60.0
LLaVA-OneVision w/ Look-M	20.0%	1.00×	57.0	58.0	62.0 / 59.8
LLaVA-OneVision w/ Ours	17.0%	0.20×	57.5	58.6	62.6 / 59.5
LLaVA-OneVision w/ Ours (Reproduce, self-port)	17.0%	0.20×	57.23	58.37	64.40 / 61.78

Table 1: PruneVid main experiment table with local reproduction rows inserted below the paper Ours rows. PLLaVA uses the released implementation; ST-LLM and LLaVA-OneVision are self-ported because the official repository marks those PruneVid backbones as not yet released.

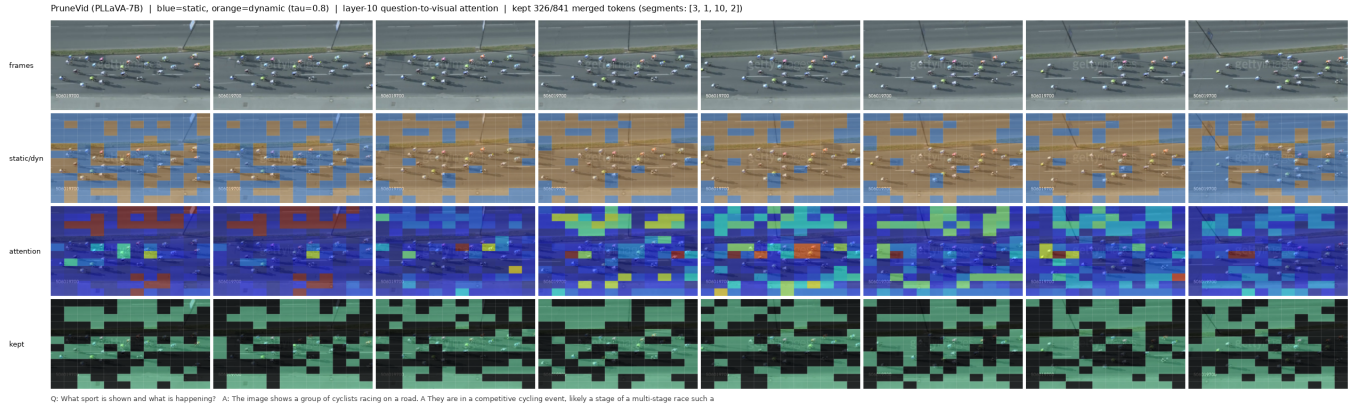


Figure 1: Qualitative visualization of PruneVid’s stages on a new cycling-race video (PLLaVA-7B, 16 frames, 12×12 grid), captured from the same instrumented code path as the table above. Row 2: static (blue) vs. dynamic (orange) patches ($\tau = 0.8$) — the moving peloton is dynamic while road and background are static and merged temporally. Row 3: layer-10 question-to-visual attention broadcast back to patches. Row 4: tokens kept by the LLM-stage top- α selection ($\alpha = 0.4$), which tracks the cyclists. More examples in [visualizations/out/](#).

2 VidCom²

Backbone	Method / Ratio	MVBench	LongVideoBench	MLVU	VideoMME	Short	Medium	Long	Average (%)
LLaVA-OV-7B	Upper bound	56.9	56.4	63.0	58.6	70.3	56.6	48.8	100.0
LLaVA-OV-7B	DyCoke, $R = 30\%$	56.6	54.7	60.3	56.1	67.1	54.6	46.6	96.5
LLaVA-OV-7B	Random, $R = 25\%$	54.2	52.7	59.7	55.6	65.4	53.0	48.3	94.8
LLaVA-OV-7B	FastV, $R = 25\%$	55.5	53.3	59.6	55.3	65.0	53.8	47.0	94.9
LLaVA-OV-7B	PDrop, $R = 25\%$	55.3	51.3	57.1	55.5	64.7	53.1	48.7	94.1
LLaVA-OV-7B	SparseVLM, $R = 25\%$	56.4	53.9	60.7	57.3	68.4	55.2	48.1	97.5
LLaVA-OV-7B	DyCoke, $R = 25\%$	49.5	48.1	55.8	51.0	61.1	48.6	43.2	87.0
LLaVA-OV-7B	VidCom², $R = 25\%$	57.2	54.9	62.5	58.6	69.8	56.4	49.4	99.6
LLaVA-OV-7B	VidCom ² , $R = 25\%$ (Reproduce)	57.00	55.27	62.09	58.33	69.33	56.56	49.11	99.3
LLaVA-OV-7B	FastV, $R = 15\%$	51.6	48.3	55.0	48.1	51.4	49.4	43.3	85.0
LLaVA-OV-7B	PDrop, $R = 15\%$	53.2	47.6	54.7	50.1	58.7	48.7	45.0	87.4
LLaVA-OV-7B	SparseVLM, $R = 15\%$	52.9	49.7	57.4	53.4	61.0	52.1	47.0	91.2
LLaVA-OV-7B	VidCom², $R = 15\%$	54.3	52.0	58.9	56.2	65.8	54.8	48.1	95.1
LLaVA-OV-7B	VidCom ² , $R = 15\%$ (Reproduce)	54.48	53.10	60.47	55.74	65.56	53.78	47.89	95.2
LLaVA-Video-7B	Upper bound	60.4	59.6	70.3	64.3	77.2	62.1	53.4	100.0
LLaVA-Video-7B	DyCoke, $R = 30\%$	57.5	55.5	60.6	61.3	73.4	59.3	51.2	93.8
LLaVA-Video-7B	FastV, $R = 25\%$	53.8	51.2	57.8	59.3	67.1	60.0	50.8	89.7
LLaVA-Video-7B	SparseVLM, $R = 25\%$	55.4	54.2	58.9	60.1	71.1	59.1	50.1	91.6
LLaVA-Video-7B	DyCoke, $R = 25\%$	50.8	53.0	56.9	56.1	65.8	53.6	48.9	86.3
LLaVA-Video-7B	VidCom², $R = 25\%$	57.0	55.5	59.0	61.7	73.0	61.7	50.0	93.6
LLaVA-Video-7B	VidCom ² , $R = 25\%$ (Reproduce)	56.88	56.62	57.24	61.85	72.89	61.67	51.00	93.5
LLaVA-Video-7B	FastV, $R = 15\%$	44.0	44.6	53.8	51.3	56.4	51.1	46.2	78.0
LLaVA-Video-7B	SparseVLM, $R = 15\%$	53.1	52.7	56.2	55.7	65.0	53.9	48.3	86.3
LLaVA-Video-7B	VidCom², $R = 15\%$	53.3	51.5	56.8	58.3	68.0	57.3	49.7	88.5
LLaVA-Video-7B	VidCom ² , $R = 15\%$ (Reproduce)	53.33	52.28	54.94	58.22	68.00	56.89	49.78	88.0

Table 2: VidCom² main experiment table with local LLaVA-OV and LLaVA-Video reproduction rows inserted below the corresponding paper rows. Average(%) follows the paper’s convention: the row’s 7-column sum (MVBench, LongVideoBench, MLVU, VideoMME, Short/Medium/Long) divided by the corresponding Upper-bound row’s 7-column sum. Reproduced Average(%) matches the paper within ≤ 0.5 pts on all four rows (OV $R=25\%$: 99.3 vs 99.6; OV $R=15\%$: 95.2 vs 95.1; Video $R=25\%$: 93.5 vs 93.6; Video $R=15\%$: 88.0 vs 88.5).

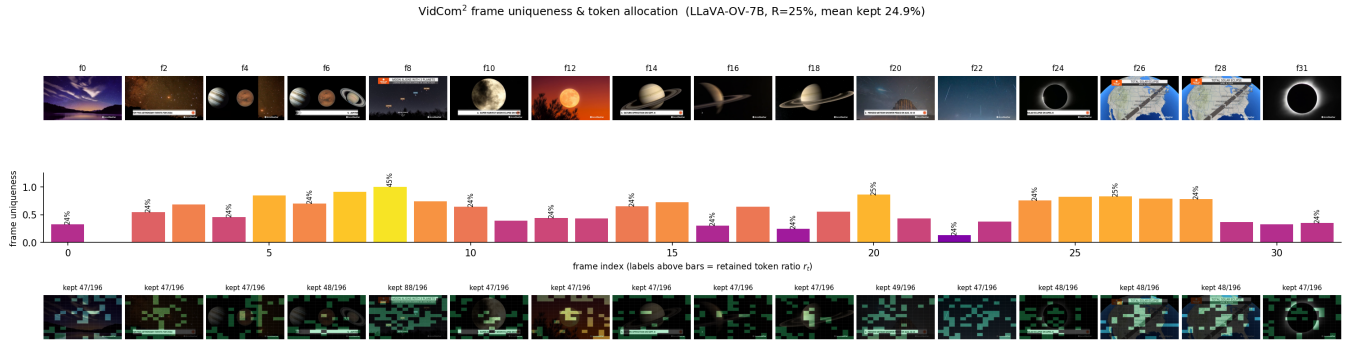


Figure 2: Qualitative visualization of VidCom² frame uniqueness and token allocation on a new VideoMME astronomy video (LLaVA-OV-7B, $R = 25\%$). Taller/brighter bars mark more unique frames, which receive a larger per-frame retention ratio r_t (labels above bars); visually redundant frames are compressed harder. Bottom: the retained 14×14 patches per frame. More examples in [visualizations/out/](#).

3 AOT

Method	Prefill FLOPs	FLOPs Ratio	Retained	MVBench	EgoSchema	LongVideoBench	VideoMME	Avg. Score	Avg. %
LLaVA-OV-7B	40.8	100%	100%	58.3	60.4	56.4	58.6	58.4	100.0
FastV	9.3	22.8%	100%	55.9	57.5	56.7	56.1	56.5	96.7
PDrop	10.5	25.7%	100%	56.1	58.0	54.1	56.4	56.2	96.2
DyCoke	8.7	21.3%	25%	53.1	59.5	49.5	54.3	54.1	92.6
VisionZip	8.7	21.3%	25%	57.9	60.3	56.5	58.2	58.2	99.7
PruneVid	8.7	21.3%	25%	57.4	59.9	55.7	57.4	57.6	98.6
FastVID	8.7	21.3%	25%	56.5	—	56.3	58.0	—	—
AOT	8.7	21.3%	25%	58.7	61.3	56.3	57.5	58.5	100.0
AOT (Reproduce)	8.7	21.3%	25%	57.75	61.18	54.75	54.89	57.14	97.8
VisionZip	7.0	17.2%	20%	57.7	59.8	55.2	57.9	57.7	98.8
PruneVid	7.0	17.2%	20%	57.2	59.7	54.7	56.9	57.1	97.8
FastVID	7.0	17.2%	20%	56.3	—	57.1	57.9	—	—
AOT	7.0	17.2%	20%	58.1	61.3	56.2	57.2	58.2	99.7
AOT (Reproduce)	7.0	17.2%	20%	57.73	61.22	54.08	54.78	56.95	97.5
VisionZip	5.2	12.7%	15%	56.5	59.8	54.4	56.1	56.7	97.1
PruneVid	5.2	12.7%	15%	56.8	59.7	55.4	56.6	57.1	97.8
FastVID	5.2	12.7%	15%	56.0	—	56.2	57.7	—	—
AOT	3.4	8.3%	15%	57.8	61.3	55.2	56.6	57.7	98.8
AOT (Reproduce)	3.4	8.3%	15%	57.33	60.98	52.80	54.22	56.33	96.5
VisionZip	3.4	8.3%	10%	53.5	58.0	49.3	53.4	53.5	91.6
PruneVid	3.4	8.3%	10%	56.2	59.8	54.5	56.0	56.6	96.9
FastVID	3.4	8.3%	10%	55.9	—	56.3	57.3	—	—
AOT	5.2	12.7%	10%	57.0	60.6	54.2	56.1	57.0	97.6
AOT (Reproduce)	5.2	12.7%	10%	56.70	60.45	52.65	53.78	55.90	95.7

Table 3: AOT main experiment table on LLaVA-OneVision with local reproduction rows inserted below the corresponding paper AOT rows. All runs follow the authors’ released evaluation config (`eval_ov-7b.sh`); EgoSchema full-test accuracy is obtained by submitting local predictions to the official validation server; LongVideoBench and VideoMME use a VTN-calibrated sweep so measured token retention matches the labeled ratios. Averages are computed over the four benchmarks, with Avg. % relative to the full-model baseline.

Method	Prefill FLOPs	FLOPs Ratio	Retained	MVBench	EgoSchema	LongVideoBench	VideoMME	Avg. Score	Avg. %
LLaVA-Video-7B	80.2	100%	100%	60.4	57.2	58.9	64.3	60.2	100.0
FastV	17.1	21.3%	100%	54.3	54.1	55.0	58.8	55.6	92.4
PDrop	19.5	24.3%	100%	55.9	54.3	54.7	61.9	56.7	94.2
VisionZip	9.3	18.9%	25%	56.7	54.7	54.7	60.7	56.7	94.2
DyCoke	9.3	18.9%	25%	50.8	—	53.0	56.9	—	—
AOT	9.3	18.9%	25%	58.8	55.4	56.2	62.4	58.2	96.7
AOT (Reproduce)	9.3	18.9%	25%	58.42	55.85 [‡]	56.24	62.52	58.26	96.8
VisionZip	9.3	11.6%	15%	56.7	54.7	54.7	60.7	56.7	94.2
AOT	9.3	11.6%	15%	57.8	55.2	55.0	62.0	57.5	95.5
AOT (Reproduce)	9.3	11.6%	15%	57.98	55.08 [‡]	55.42	62.11	57.65	95.8

Table 4: AOT main experiment table on LLaVA-Video with local reproduction rows inserted below the corresponding paper AOT rows. [‡]EgoSchema full-test labels are not public; accuracy is obtained by submitting the local predictions to the official validation server. Averages are computed over the four benchmarks, with Avg. % relative to the full-model baseline.

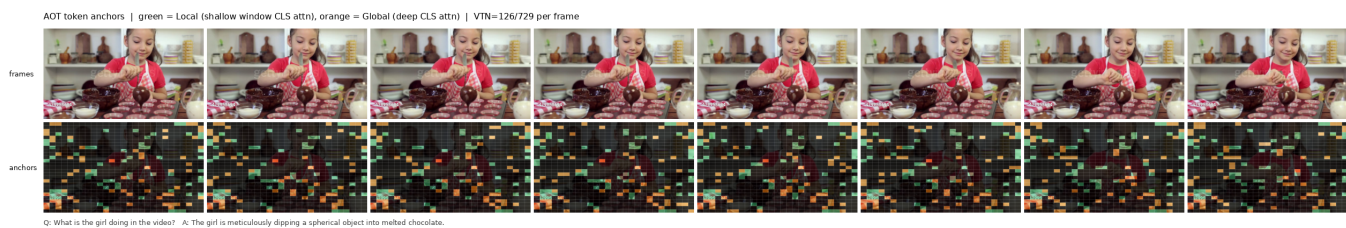


Figure 3: Qualitative visualization of AOT’s local/global token anchors on a new cooking video (LLaVA-OV-7B, VTN= 126/729 per frame), captured from the same instrumented code path as the tables above. Top: sampled frames; bottom: green = Local anchors (shallow-layer window CLS attention), orange = Global anchors (deep-layer CLS attention); unselected patches are dimmed and later merged into the anchors via optimal transport. Anchors concentrate on the girl’s face, hands, and the chocolate she is dipping. More examples in [visualizations/out/](#).